

An Autonomous Internet of Things (IoT) pipeline network for Image acquisition in Agricultural Applications

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Abstract— Agriculture is considered to be a self-sufficient component of the country's economy. In terms of agricultural output, India is second only to the United States in terms of production. Irrigation, fertiliser, and crop rotation are three of the most significant elements in agricultural production. The usage of the Internet of Things can make crop production predictions and other components that contribute to high yielding crops more accurate and efficient (IoT). In the past, a pipeline for making predictions was created. A solution based on the Internet of Things (IoT) is being developed and designed to address this problem (IoT). When operating in a Wireless Sensor Network (WSN), sensor nodes are required to perform tasks such as observation, data collection, and transaction processing in order to exhaust their energy. Farmers saving time and money by collecting this information on their own is a win-win situation. To make advantage of this network, you must have a wireless sensor network (WSN), which must be capable of transmitting large amounts of data at a low rate while also consuming little power and communicating over short distances.

Keywords— Internet of Things, wireless sensor network, Image acquisition, Agricultural Applications.

I. INTRODUCTION

New approaches that can feed nine billion people by 2050 have been greatly developed thanks to the Internet of Things' role in Precision Agriculture. In the modern day, farmers use real-time data from sensors and monitor their fields to increase yields and profitably gather more crops. Sensors are being used in farming areas all over the developed world to automate farming tasks based on the current state of the environment [1]. As a result of the country's unique climatic circumstances, India is the second-largest producer of agricultural products in the world. With IoT-based precision agriculture and other recent technical breakthroughs, we may expect a long-term increase in agricultural output. Farming decisions can be made more accurately thanks to the Internet of Things (IoT) in precision farming [2].

Many sensor nodes are coupled in Wireless Sensor Networks (WSNs), restricting processing power. As more sensor data is sent to the sink node, the network becomes more complex. Data processing is a major issue in the WSN,

which has a significant impact on the network's overall performance. Sensor nodes are often powered by batteries and have a limited amount of energy [3]. Much data is collected and analysed in precision agriculture.

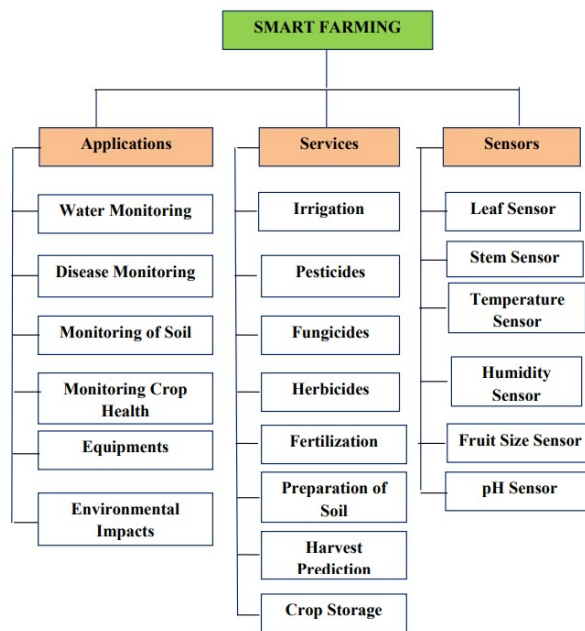


Fig. 1. Smart forming using Machine learning

With the use of WSN, precision agriculture can help farmers improve their crop yields by replacing current agricultural methods with newer technology. Machine learning is the ability to learn from experience [4]. Based on the predefined equations, the algorithms learn from the input data and progressively train the samples, so improving their overall performance. It aids in decision-making with minimal human intervention based on real-world events. In today's world, machine learning algorithms may be found in almost every field of study. Accuracy, on the other hand, is affected by the quality of the data. Thus, dataset representations and target variables play a crucial role in machine learning techniques. As a new ML technique for quantifying and understanding big data applications, WSN and IoT enabled

smart farming is being developed [5]. Figure 1 depicts the numerous ways in which machine learning is used in intelligent agriculture.

II. IOT PIPELINE NETWORK FOR IMAGE ACQUISITION

An image processing system begins with image acquisition. Image acquisitions are typically used to convert real-world data into a collection of numerical data that can be further processed by a computer. Sensor nodes that are able to capture images are used. Pixels are two-dimensional, minuscule points that make up an image in two dimensions. Here, the utility of two already-existing variables is being discussed [6]. Image processing is a technique for enhancing and making use of digital images, and it can be employed to gather useful data. A digital image can be described as a collection of statistics illustrating the spatial distribution of certain field parameters, such as the reflectance of electromagnetic radiation, emissivity, temperature, or any geophysical elevation, in the broadest sense.

Images can be improved through image processing, which is the act of enhancing images that have been captured by various equipment such as a camera or satellites, aircrafts, or even in everyday life [7]. Processing techniques have developed a wide range of methodologies during the last few decades. Satellite image enhancement techniques based on parallel processing have been implemented as the primary contribution of this work. In this approach, parallel image processing is proposed as a pipeline, with a primary focus on the current efficacy of remote sensing image processing [8].

Most commonly, remote sensing refers to the practise of photographing an object without the sensor's unwavering touch with it. It is possible to construct satellite-based remote sensing with numerous satellites that can be used to collect data from the ocean, ice, and nearby areas. Since mapping and ecological monitoring are well-established methods for gathering accurate data at a cheap cost, they are worthy of consideration [9]. Using a computer system, the digital photos can be processed to depict satellite images, giving rise to the term "image processing method." The BS receives data from the sensor nodes over many hops. Otherwise, this data is recorded in a database and made available to the general public.

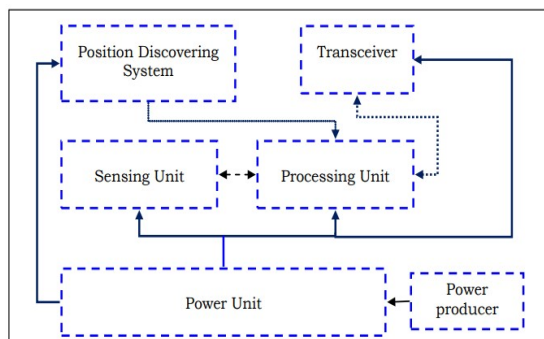


Fig. 2. WSN Architecture block diagram

The WSN's sensor nodes are frequently the most important gadgets since they assist monitor an area that is normally off-limits to social interaction [10]. As shown in Figure 2, the construction of a sensor node can be characterised using basic blocks such as memory storage, sensing units, and transducers.

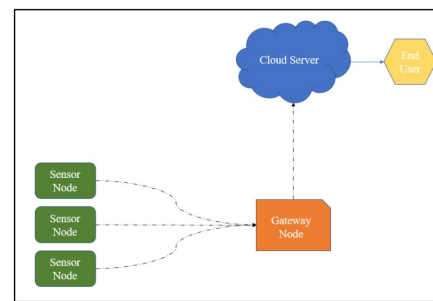


Fig. 3. Structure of IoT Network

Many other kinds of IOT-ready applications might be derived from image-processing-based sensors and the data they create [11]. For a quick look at image processing's place in the Internet of Things, this paper examines a number of different IOT use cases.

Using web-connected devices to automate common or simple tasks is referred to as the Internet of Things (IoT). Wireless sensors and gadgets that collect and send data to the internet are what we're describing in this scenario. A computer vision method or a machine learning model, for example, could use image and analogue sensor data as inputs. It is common for an Internet of Things (IoT) system to have four primary components: IoT devices; communications technology the internet; and data storage. Connected networks are shown in Figure 3. Using an IoT system to communicate in remote and difficult-to-reach locations is a major advantage for farmers around the world [12].

III. METHODOLOGY

In the context of digital agriculture, these devices may be used to monitor soil moisture, nutrient quality, or microbial activity [6, 7]. The WSN captures and sends sensor data to the cloud, where it can be further analysed and, in some cases, used for closed-loop control. The Internet of Things Wireless Sensor Networks (WSNs) for industrial lyophilization and smart agriculture are the subject of this research.

Hand-held devices with Internet of Things capabilities are created for rural farmers, making them simple to use and portable. People can view information on agricultural fields on the device's integrated graphical display [13]. Farmers can monitor their fields from any location owing to the combination of this handheld gadget with LoRa and Wi-Fi technologies. The figure 4 depicts a handheld gadget that is based on the Internet of Things, and it highlights the many components that make it up. The sensor mote and gateway are created and customised using this approach, which is the major method. As part of our research, we are developing a bespoke sensor mote, gateway, and handheld device. Customization of the prototype is a typical technique for lowering costs while simultaneously assuring a robust and straightforward system [14].

The process of customising each component begins with the selection of each component. In many cases, the power consumption metric is taken into consideration when selecting a sensor mote component since the sensor mote is placed in agricultural areas where access to an external power source is not readily available.

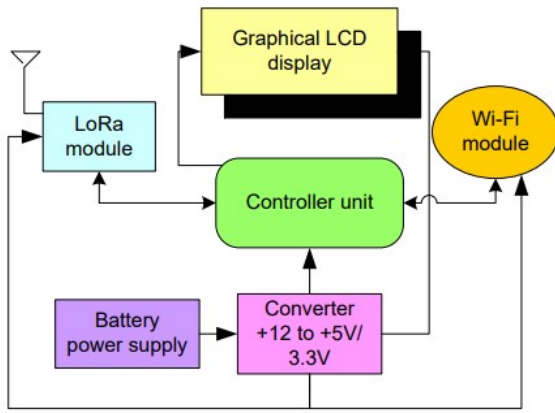


Fig. 4. IoT based device architecture

When transmitting data, the sensor mote's power consumption is greatly reduced by using wireless communication.

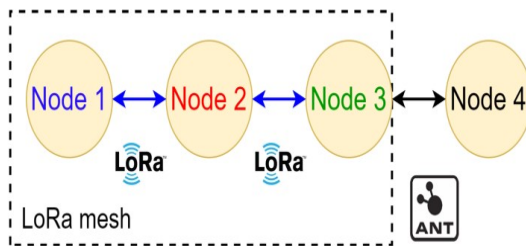


Fig. 5. Network Setup for Experiment

One node's power consumption was tested in real-world conditions for the network depicted in Fig 5. HDC2010 temperature and humidity sensors are also connected to each sensor. Each node will broadcast a 64-byte packet including a temperature and humidity reading for each cycle [15]. The energy consumption of a mesh network changes based on the structure of the network and the position of the participating nodes in the hierarchy. The proposed LoRa mesh protocol connects nodes 1, 2, and 3 as depicted in Figure 5. To interact with node 4, ANT is used by node 3. Data from nodes 4, 3, and 2 (ANT and LoRa) are to be routed to node 1 via this setup (LoRa). Node 4 transmits one package via ANT to node 3 for each TDMA scheduling cycle, and node 3 forwards this package and its own package to node 2 through LoRa for each TDMA scheduling cycle. Node 2 will then send three packages to Node 1: two received and one generated. This test uses the same LoRa configuration as the field deployments, with a transmission power of +18 dBm.

LoRa (long-range) communication is used by the gateway node to connect to the cloud. It receives data from all of the sensor nodes and sends it to a cloud server over an internet network in the neighbourhood. Gateway and internet connection are separate, requiring an additional step in data transfer. The ThingSpeak cloud server, a basic analytics tool for IoT projects, receives the data from the internet router. Figure 6 depicts the IoT system under discussion.

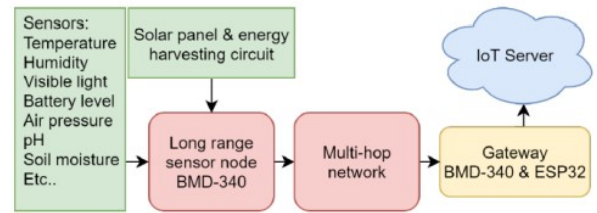


Fig. 6. IoT based complete system Architecture

IV. RESULTS

As a result, a low-power wireless communication module is selected for long-range data transmission. The SX 1278 LoRa module is the one we've chosen.

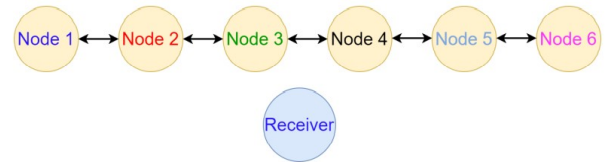


Fig. 7. Network Pipeline structure

As shown in Figure 8(a), all nodes except node 4 demonstrate greater than 99 percent PDRi after a one-week test. Total PDRd in Figure 8(b) illustrates that the network's stability is quite robust and does not change over time.

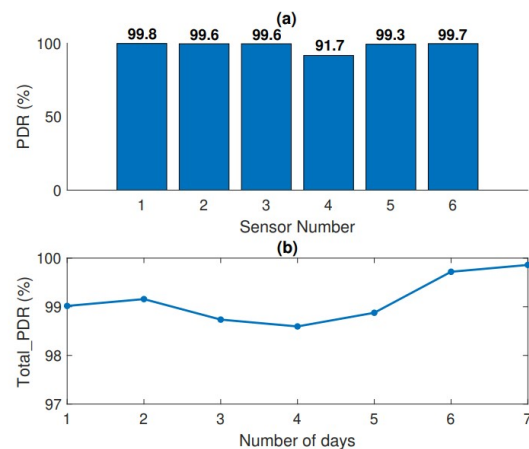


Fig. 8. PDR, Total_PDR for the stability test

Table 1 illustrates the amount of water that has been used for irrigation purposes over the years. In order to maintain the health of the plants, it is necessary to maintain a steady level of water. Following the procedure outlined above, the first irrigation was carried out on February 1st at 12 noon. The threshold value for this process is always the same. According to this table, there is a 2500ml difference in water use between the planned and automatic forms of irrigation, which is significant. The findings of the water consumption study, which was done over the course of the designated cultivation days, are summarised in Table 3.4. The ARV Recommendation System suggests cultivating at a moisture content greater than or equal to 35Volumetric Water Content while growing a variety of crops (threshold value). The irrigation pump will be activated if the moisture level exceeds a certain threshold.

Table I. Analysis of water consumption during the cultivation

Mode of Cultivation	No of Days	Water consumption per a day	Total water consumption
Sheduled	5	600	3000
Automaticce	5	250	1250

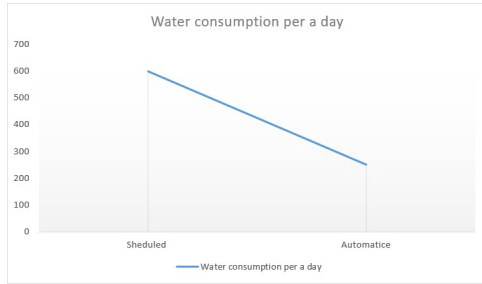


Fig. 9. comparision graph for water consumption per day

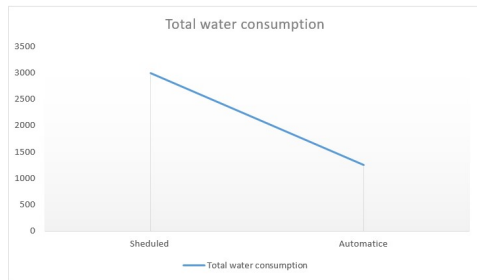


Fig. 10. comparision graph for water consumption per day

V. CONCLUSION

While the goals and requirements of the agriculture business are accomplished, the Internet of Things system presented is not limited to this area. Wireless sensor networks (WSN) can be utilised for a variety of applications in which a high data rate and low power consumption are required while delivering a large amount of data at low frequency interstices are required. As the Internet of Things (IoT) continues to develop, edge computing, machine learning, and computer vision will become increasingly crucial technologies to consider. Crop monitoring and yield prediction are two of the best examples of this new generation of Internet of Things systems, which are moving away from the low data rate, restricted bandwidth models of the past to more advanced models. According to data collected throughout the trial period, daytime temperatures are high and humidity is low, however nighttime temperatures are low and humid. During the day, the rate of evaporation of water also increases significantly. The relative humidity during the day will be lower than the relative humidity during the night, decreasing the threshold value. After that, farming advise will be able to predict the crop and the highest possible yield in a specific region.

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